

Finding Resonance: Standing Waves on a String

SimuPhysics Standing Waves Simulation · SimuPhysics

Years 8-9	~30 minutes	Science · Physics
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NOTICE — AI-generated worksheet

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Simulation: <https://simuphysics.com/view/oscillations-and-waves-standing-waves-simulation.html>

Curriculum links: AC9S9U04 · AC9S8U05

Learning intentions

- I can identify nodes and antinodes in a standing wave pattern.
- I can relate the frequency of vibration to the number of nodes and antinodes produced.
- I can describe standing waves as an example of energy transfer through a medium.

Getting started

Open the SimuPhysics Standing Waves Simulation. Set the string tension to a fixed value and start with a low frequency, gradually increasing it while watching the string's motion.

Tasks

1. At a low frequency, describe what the string looks like. Can you count any clear nodes or antinodes yet?

2. Slowly increase the frequency until you find a pattern with one clear 'loop', or antinode, between the fixed ends. Record the approximate frequency.

3. Find the frequencies that produce 1, 2, 3 and 4 loops and record the number of nodes for each.

Number of loops (antinodes)	Approx. frequency	Number of nodes

4. Describe the pattern you notice between the number of loops and the frequency needed to produce them.

5. Increase the string tension and find the frequency that produces the same number of loops as before. What happened to the frequency needed?

6. Explain what a standing wave pattern tells you about energy transfer along the string, including whether the energy appears to move from one end to the other.

7. Identify the positions of the nodes in your 2-loop pattern. Why do these points stay still while the rest of the string moves?

Extension (optional)

Musical instruments like guitars use standing waves on strings. Use what you found about tension and frequency to explain why tightening a guitar string raises the pitch of the note it produces.
